

# Notation

This page is a reference, not a section: there is nothing to work through here, and it carries no exercises and no quick checks. Use it when you meet a symbol and want to know what it means and where the book fixed it. The symbols are grouped by the kind of thing they name, not by chapter, because a reader arrives with a symbol and not with a chapter number. Every entry links to the place where the book defines it; follow the link for the hypotheses, which a one-line gloss cannot carry.

Read the conventions first. Most of what looks like a clash between two symbols is resolved by one of them.

## 23.-1.1 The standing conventions

**The field.**  $F$  is an arbitrary field throughout, and  $\mathbb{F}_p$  is the field with  $p$  elements (Definition 0.7.1). Chapters 10 to 12 and much of Part V need  $\mathbb{R}$  or  $\mathbb{C}$ , and say so where they do. Vector spaces are  $V, U, W$ , plain italic, and are finite-dimensional unless the statement says otherwise.

**Bold is for matrices and vectors, never for maps or spaces.** A matrix is a bold upright capital,  $\mathbf{A}$ ; a vector is a bold lowercase letter,  $\mathbf{v}$ ; an operator or linear map is plain italic,  $T$ ; a space is plain italic,  $V$ . Entries stay plain:  $a_{ij}$ , or  $(\mathbf{A})_{ij}$ . A Greek letter naming a matrix is bold too, as  $\mathbf{\Lambda}$  or  $\mathbf{\Sigma}$ .

**Maps and composition.** Linear maps are  $T, S: V \rightarrow W$ , never  $f$ . Composition is written  $ST$ , and it means: first  $T$ , then  $S$ . For functions in general the book writes  $g \circ f$ .

**Coordinates.**  $[\mathbf{v}]_{\mathcal{B}}$  is the coordinate vector of  $\mathbf{v}$  in the ordered basis  $\mathcal{B}$  (Definition 1.5.2), and  $[T]_{\mathcal{B}}^{\mathcal{C}}$  is the matrix of  $T$  with input basis  $\mathcal{B}$  and output basis  $\mathcal{C}$  (Definition 3.6.1). The subscript is the basis read into the first slot and the superscript the basis read into the second: for a map that is the input basis below and the output basis above, and for a bilinear form  $[\beta]_{\mathcal{B}}^{\mathcal{B}}$  it is the basis used on the first argument below and the one used on the second above. Both slots are always written out, even when they hold the same basis. A basis is an **ordered** list, written in parentheses.

**Transpose and adjoint.**  $\mathbf{A}^{\top}$  is the transpose (Definition 0.9.10) and  $\mathbf{A}^*$  the conjugate transpose (Definition 10.1.2), which is also the symbol for the adjoint of an operator (Definition 10.6.1). The prime  $T'$  is the dual map (Definition 4.4.1), and never a derivative of a map.

**Inner products are linear in the first slot** and conjugate-linear in the second (Definition 10.1.1). Over  $\mathbb{R}$  the distinction disappears.

**Ordering of eigenvalues and singular values.** For Hermitian  $\mathbf{A}$ , the eigenvalues are listed **decreasing**,  $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_n(\mathbf{A})$ , with multiplicity. Singular values are listed decreasing as well,  $\sigma_1(\mathbf{A}) \geq \sigma_2(\mathbf{A}) \geq \cdots \geq 0$  (Definition 12.8.1). Where a proof wants the increasing order it says so.

**Indices start at 1**, with three announced exceptions: Chapter 11 §09, where circulant and Fourier subscripts run from 0 to  $n-1$  and are read modulo  $n$ ; Chapter 21 §§06 to 10, where the coordinates of  $F^{n+1}$  run from 0 so that  $\mathbb{P}^n(F)$  has dimension  $n$  (Definition 21.6.2); and Chapter 22 §09, which follows Chapter 11 §09 for group elements and characters.

**Defining, not asserting.**  $:=$  means “is defined to be”. A plain  $=$  asserts an equality that has

been or will be proved.

## 23.-1.2 Numbers, sets and fields

| Symbol                                                       | Meaning                                                                  | Defined at                                           |
|--------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------|
| $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ | natural numbers (with 0), integers, rationals, reals, complex numbers    | Definition 0.3.1                                     |
| $F, \mathbb{F}_p$                                            | a field; the field with $p$ elements                                     | Definition 0.7.1                                     |
| $\text{char } F$                                             | characteristic of a field: the least $n > 0$ with $n \cdot 1 = 0$ , or 0 | Definition 0.7.2                                     |
| $\bar{z},  z $                                               | complex conjugate and modulus                                            | Definition 0.6.2                                     |
| $\{x \in S : P(x)\}$                                         | set-builder notation, with a colon                                       | Definition 0.3.2                                     |
| $\emptyset, S \subseteq T$                                   | the empty set; inclusion                                                 | Definition 0.3.3, Definition 0.3.4                   |
| $S \cup T, S \cap T, S \setminus T, S^c$                     | union, intersection, difference, complement                              | Definition 0.3.6, Definition 0.3.7, Definition 0.3.8 |
| $S \times T, (x_1, \dots, x_n)$                              | Cartesian product and $n$ -tuples                                        | Definition 0.3.10, Definition 0.3.11                 |
| $\mathcal{P}(S)$                                             | power set, the set of all subsets                                        | Definition 0.3.12                                    |
| $f: X \rightarrow Y,$<br>$x \mapsto f(x)$                    | a function and its rule                                                  | Definition 0.4.1                                     |
| $f(A), f^{-1}(B)$                                            | image of a subset, preimage of a subset                                  | Definition 0.4.2                                     |
| $g \circ f, \text{id}_X, f^{-1}$                             | composition, identity function, inverse function                         | Definition 0.4.4, Definition 0.4.5, Definition 0.4.7 |
| $x \sim y, [x], X/\sim$                                      | equivalence relation, equivalence class, quotient set                    | Definition 0.5.2, Definition 0.5.3, Definition 0.5.5 |
| $S_n, \text{sgn}(\sigma), \text{inv}(\sigma)$                | symmetric group; sign of a permutation; number of inversions             | Example (Groups), Definition 6.3.2, Definition 6.3.1 |
| $A_n$                                                        | alternating group, the even permutations                                 | Definition 6.3.3                                     |
| $F[x], F[x]_{\leq n}$                                        | polynomials over $F$ ; those of degree at most $n$                       | Definition 0.8.2, Definition 0.8.4                   |
| $\deg p, p(c)$                                               | degree; evaluation at a scalar                                           | Definition 0.8.3, Definition 0.8.6                   |
| $p \mid q, \gcd(p, q), \text{lcm}(p, q)$                     | divisibility, greatest common divisor, least common multiple in $F[x]$   | Definition 5.1.1, Definition 5.2.2, Definition 5.2.4 |
| $\langle p \rangle$                                          | the ideal of $F[x]$ generated by $p$ : all multiples of $p$              | Definition 5.2.1                                     |
| $\text{mult}_c(f)$                                           | multiplicity of $c$ as a root of $f$                                     | Definition 5.4.1                                     |
| $p'$                                                         | formal derivative of a polynomial                                        | Definition 5.4.3                                     |
| $\ell_i$                                                     | Lagrange basis polynomial, $\ell_i(c_j) = \delta_{ij}$                   | Definition 5.6.1                                     |

## 23.-1.3 Spaces

| Symbol                                              | Meaning                                                                                  | Defined at                                    |
|-----------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------|
| $V, U, W$                                           | vector spaces over $F$ ; $U$ is usually a subspace                                       | Definition 1.1.1,<br>Definition 1.2.1         |
| $F^n, F^S$                                          | column vectors; <b>all</b> functions $S \rightarrow F$                                   | Definition 1.1.1                              |
| $F^{(S)}$                                           | the free vector space on $S$ : the <b>finitely supported</b> functions $S \rightarrow F$ | Definition 14.2.3                             |
| $M_{m \times n}(F), M_n(F)$                         | matrices over $F$                                                                        | Definition 0.9.1                              |
| $\text{span}(S)$                                    | span; the span of the empty set is $\{\mathbf{0}\}$                                      | Definition 1.3.2                              |
| $\dim V, \dim_F V$                                  | dimension, over $F$ when the field matters                                               | Definition 1.6.1                              |
| $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ | an <b>ordered</b> basis; $\mathcal{E}$ is the standard basis                             | Definition 1.5.1,<br>Example (Standard Bases) |
| $[\mathbf{v}]_{\mathcal{B}}$                        | coordinate vector of $\mathbf{v}$ in $\mathcal{B}$                                       | Definition 1.5.2                              |
| $U + W, U \oplus W$                                 | sum of subspaces; internal direct sum                                                    | Definition 1.7.1,<br>Definition 1.7.2         |
| $V \times W$                                        | product of vector spaces (the external direct sum)                                       | Definition 3.9.1                              |
| $\mathbf{v} + U, V/U$                               | coset; quotient space                                                                    | Definition 3.9.2,<br>Definition 3.9.3         |
| $\text{codim } U$                                   | codimension, $\dim V/U$                                                                  | Definition 4.5.1                              |
| $V^*, \varphi$                                      | dual space; a linear functional, always a Greek letter                                   | Definition 4.1.2,<br>Definition 4.1.1         |
| $\varphi_1, \dots, \varphi_n$                       | the dual basis of $\mathcal{B}$                                                          | Definition 4.1.3                              |
| $U^0$                                               | annihilator of $U$ inside $V^*$                                                          | Definition 4.2.1                              |
| $V^{**}, \text{ev}_{\mathbf{v}}$                    | double dual; evaluation at $\mathbf{v}$                                                  | Definition 4.3.1                              |
| $V_{\mathbb{C}}$                                    | complexification of a real space                                                         | Definition 1.9.2                              |
| $\mathcal{F}$                                       | a family of operators in $\mathcal{L}(V)$                                                | Definition 8.9.1                              |

## 23.-1.4 Vectors, matrices and their operations

| Symbol                                                       | Meaning                                                            | Defined at                              |
|--------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------|
| $\mathbf{v}, \mathbf{u}, \mathbf{w}, \mathbf{x}, \mathbf{y}$ | vectors: bold lowercase                                            | Definition 1.1.2                        |
| $\mathbf{0}, \mathbf{1}$                                     | the zero vector; the all-ones vector                               | Definition 1.1.2,<br>Definition 8.11.1  |
| $\mathbf{e}_1, \dots, \mathbf{e}_n$                          | the standard basis of $F^n$                                        | Example (Standard Bases)                |
| $\mathbf{E}_{ij}$                                            | matrix unit: 1 in entry $(i, j)$ , zeros elsewhere                 | Example (Standard Bases)                |
| $\mathbf{A}, \mathbf{B}, \mathbf{C}, a_{ij}$                 | matrices, bold upright capitals; their entries, plain              | Definition 0.9.1                        |
| $\mathbf{0}_{m \times n}, \mathbf{I}, \mathbf{I}_n$          | zero matrix, plain (bold $\mathbf{0}$ as a block); identity matrix | Definition 0.9.2,<br>Definition 0.9.3   |
| $\delta_{ij}$                                                | Kronecker delta: 1 when $i = j$ , else 0                           | Definition 0.9.3                        |
| $\text{diag}(d_1, \dots, d_n)$                               | the diagonal matrix with these diagonal entries                    | Definition 0.9.4                        |
| $\mathbf{A}^{\top}, \mathbf{A}^*$                            | transpose; conjugate transpose                                     | Definition 0.9.10,<br>Definition 10.1.2 |

| Symbol                                                                        | Meaning                                                                                                                     | Defined at                                                    |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| $\text{tr } \mathbf{A}$                                                       | trace, the sum of the diagonal entries                                                                                      | Definition 0.9.12                                             |
| $\det \mathbf{A}$                                                             | determinant                                                                                                                 | Definition 6.4.1                                              |
| $\text{adj } \mathbf{A}$                                                      | adjugate, the transposed matrix of cofactors                                                                                | Definition 6.6.2                                              |
| $M_{ij}, C_{ij}$                                                              | minor and cofactor $(-1)^{i+j} M_{ij}$                                                                                      | Definition 6.6.1                                              |
| $\mathbf{A}^{-1}$                                                             | inverse of a non-singular matrix                                                                                            | Definition 0.9.13                                             |
| $p(\mathbf{A})$                                                               | a polynomial evaluated at a matrix                                                                                          | Definition 0.9.14                                             |
| $\mathbf{P}_{ij}, \mathbf{D}_i(c),$<br>$\mathbf{I} + c\mathbf{E}_{ij}$        | the three elementary matrices: swap, scale, add                                                                             | Definition 2.4.1                                              |
| $\mathbf{P}_\sigma$                                                           | permutation matrix                                                                                                          | Definition 2.4.2                                              |
| $\mathbf{A} = \mathbf{L}\mathbf{U},$<br>$\mathbf{PA} = \mathbf{L}\mathbf{U}$  | the LU and PLU factorizations                                                                                               | Definition 2.6.1,<br>Definition 2.6.2                         |
| $\text{col}(\mathbf{A}), \text{row}(\mathbf{A}),$<br>$\text{nul}(\mathbf{A})$ | column space, row space, null space                                                                                         | Definition 2.5.1,<br>Definition 2.5.2,<br>Definition 2.5.3    |
| $\text{rank } \mathbf{A}, \text{nullity } \mathbf{A}$                         | rank and nullity of a matrix                                                                                                | Definition 2.5.4,<br>Definition 2.5.5                         |
| $\mathbf{A} \sim \mathbf{B}$                                                  | similar: $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$                                                                 | Definition 3.7.2                                              |
| $\mathbf{A} \simeq \mathbf{B}$                                                | congruent: $\mathbf{B} = \mathbf{P}^\top \mathbf{A} \mathbf{P}$                                                             | Definition 13.2.1                                             |
| $\mathbf{A} \approx \mathbf{B}$                                               | equivalent over $F[x]$ : $\mathbf{B} = \mathbf{U}\mathbf{A}\mathbf{V}$ with $\mathbf{U}, \mathbf{V}$ invertible over $F[x]$ | Definition 9.7.2                                              |
| equivalent<br>matrices                                                        | $\mathbf{B} = \mathbf{Q}\mathbf{A}\mathbf{P}$ with $\mathbf{P}, \mathbf{Q}$ non-singular                                    | Definition 3.8.1                                              |
| $\mathbf{A} \oplus \mathbf{B}$                                                | block diagonal matrix with the given square diagonal blocks                                                                 | Definition 7.1.2                                              |
| $\mathbf{M}/\mathbf{A}, \mathbf{M}/\mathbf{D}$                                | Schur complement of an invertible block of a block matrix                                                                   | Definition 7.3.1                                              |
| $\mathbf{A} \otimes \mathbf{B}$                                               | Kronecker product                                                                                                           | Definition 7.4.1                                              |
| $\text{vec } \mathbf{X}$                                                      | vectorization: the columns of $\mathbf{X}$ stacked into one column                                                          | Definition 7.4.2                                              |
| $\mathbf{A} \circ \mathbf{B}$                                                 | Hadamard (entrywise) product                                                                                                | Definition 12.7.1                                             |
| $[\mathbf{A}, \mathbf{B}] = \mathbf{AB} - \mathbf{BA}$                        | commutator                                                                                                                  | Definition 7.5.1                                              |
| $\mathbf{J}$                                                                  | the all-ones matrix, every entry 1                                                                                          | Example<br>(Orthogonally<br>diagonalizing a 3 by 3<br>matrix) |
| $\mathbf{A}'(t), \mathbf{x}'(t)$                                              | entrywise derivative of a matrix- or vector-valued function of a real variable                                              | Definition 15.6.1                                             |
| $e^{\mathbf{A}}, f(\mathbf{A})$                                               | matrix exponential; a function of a matrix through its Jordan form                                                          | Definition 9.9.2,<br>Definition 9.9.1                         |
| $\lim_m \mathbf{A}^m$                                                         | entrywise convergence of matrices                                                                                           | Definition 9.10.1                                             |
| $\mathbf{Ax} = \mathbf{b}, \mathbf{Ax} = \mathbf{0}$                          | a linear system and its homogeneous version                                                                                 | Definition 2.1.1,<br>Definition 2.1.2                         |
| echelon form                                                                  | the staircase shapes elimination produces                                                                                   | Definition 2.2.3,<br>Definition 2.2.4                         |

## 23.-1.5 Maps and their spaces

| Symbol                                              | Meaning                                                                                                   | Defined at                              |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------|
| $T, S: V \rightarrow W$                             | linear maps; $ST$ means first $T$ , then $S$                                                              | Definition 3.1.1                        |
| $T_{\mathbf{A}}$                                    | the map $F^n \rightarrow F^m$ , $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$                                 | Example<br>(Multiplication by a Matrix) |
| $\ker T$ , $\operatorname{im} T$                    | kernel and image                                                                                          | Definition 3.2.1,<br>Definition 3.2.2   |
| $\operatorname{rank} T$ , nullity $T$               | $\dim \operatorname{im} T$ and $\dim \ker T$                                                              | Definition 3.3.2,<br>Definition 3.3.1   |
| $\mathcal{L}(V, W)$ , $\mathcal{L}(V)$              | all linear maps $V \rightarrow W$ ; the operators on $V$                                                  | Definition 3.4.1                        |
| $\operatorname{End}(V)$                             | the same space $\mathcal{L}(V)$ , named as an <b>algebra</b> under composition                            | Definition 22.1.1                       |
| $p(T)$ , $\operatorname{id}_V$                      | a polynomial in an operator; the identity operator                                                        | Definition 3.4.2,<br>Definition 3.1.1   |
| $[T]_{\mathcal{B}}^{\mathcal{C}}$                   | matrix of $T$ from the input basis $\mathcal{B}$ to the output basis $\mathcal{C}$                        | Definition 3.6.1                        |
| $[\operatorname{id}]_{\mathcal{B}}^{\mathcal{C}}$   | change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$                                          | Definition 3.7.1                        |
| $V \cong W$                                         | isomorphic vector spaces                                                                                  | Definition 3.5.2,<br>Definition 3.5.3   |
| $T'$                                                | dual map $W^* \rightarrow V^*$ ; the star is kept for adjoints                                            | Definition 4.4.1                        |
| $\operatorname{tr} T$ , $\det T$                    | trace and determinant of an operator                                                                      | Definition 3.10.2,<br>Definition 6.5.1  |
| $T _U$ , $\bar{T}$                                  | restriction of $T$ to an invariant subspace; the operator it induces on $V/U$                             | Definition 8.1.1,<br>Definition 8.1.2   |
| $\operatorname{GL}_n(F)$ , $\operatorname{SL}_n(F)$ | general and special linear groups                                                                         | Example (Groups),<br>Theorem 6.5.9      |
| $\mathcal{M}(V_1, \dots, V_k; W)$                   | multilinear maps $V_1 \times \dots \times V_k \rightarrow W$ ; the semicolon separates inputs from output | Definition 14.1.2                       |

## 23.-1.6 Inner products and norms

| Symbol                                                                                | Meaning                                                                                      | Defined at                              |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------|
| $\langle \mathbf{u}, \mathbf{v} \rangle$                                              | inner product: <b>linear in the first slot</b> , conjugate-linear in the second              | Definition 10.1.1                       |
| $\langle \mathbf{A}, \mathbf{B} \rangle = \operatorname{tr}(\mathbf{B}^* \mathbf{A})$ | the Frobenius inner product on matrices                                                      | Definition 10.1.1                       |
| $\ \mathbf{v}\ $                                                                      | norm; the induced norm $\sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$ unless said otherwise | Definition 10.1.3,<br>Definition 15.1.1 |
| $\mathbf{u} \perp \mathbf{v}$ , $\theta$                                              | orthogonal vectors; the angle between two non-zero real vectors                              | Definition 10.1.4,<br>Definition 10.1.5 |
| $\mathbf{G}$                                                                          | Gram matrix of a list, $(\mathbf{G})_{ij} = \langle \mathbf{v}_j, \mathbf{v}_i \rangle$      | Definition 10.2.3                       |
| $U^\perp$                                                                             | orthogonal complement                                                                        | Definition 10.3.1                       |
| $P_U$ , $d(\mathbf{v}, U)$                                                            | orthogonal projection onto $U$ ; distance to $U$                                             | Definition 10.3.2,<br>Definition 10.3.3 |
| $T^*$                                                                                 | adjoint of an operator                                                                       | Definition 10.6.1                       |

| Symbol                                                                                | Meaning                                                                                                             | Defined at                               |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| self-adjoint,<br>normal                                                               | $T^* = T$ ; $T^*T = TT^*$                                                                                           | Definition 10.6.2,<br>Definition 10.6.3  |
| $O(n)$ , $SO(n)$ , $U(n)$                                                             | orthogonal, special orthogonal and unitary groups                                                                   | Definition 10.7.3                        |
| $\mathbf{H}_w$ , $\mathbf{G}(i, j; c, s)$                                             | Householder reflection; Givens rotation                                                                             | Definition 10.8.1,<br>Definition 10.8.2  |
| $P_n$ , $T_n$                                                                         | Legendre and Chebyshev polynomials                                                                                  | Definition 10.10.1,<br>Definition 23.7.4 |
| $\ \mathbf{x}\ _p$                                                                    | the $p$ -norm; $q$ is the conjugate exponent,<br>$1/p + 1/q = 1$                                                    | Definition 15.1.1,<br>Corollary 17.11.5  |
| $\ f\ _\infty$                                                                        | sup norm on $C[0, 1]$                                                                                               | Definition 15.1.1                        |
| $B_{\ \cdot\ }$ , $B_1$ , $B_2$ , $B_\infty$                                          | closed unit ball of a norm; the balls of the three<br>$p$ -norms                                                    | Definition 15.1.2                        |
| $\ \mathbf{A}\ _F$                                                                    | Frobenius norm, $\sqrt{\text{tr}(\mathbf{A}^*\mathbf{A})}$ ; the subscript is<br>obligatory                         | Lemma 10.7.5                             |
| $\ \mathbf{A}\ $ , $\ \mathbf{A}\ _1$ , $\ \mathbf{A}\ _\infty$                       | operator norm induced by a vector norm; largest<br>column sum; largest row sum                                      | Definition 15.3.2                        |
| $\kappa(\mathbf{A}) =$<br>$\ \mathbf{A}\  \ \mathbf{A}^{-1}\ $                        | condition number                                                                                                    | Definition 15.8.1                        |
| $\ \mathbf{y}\ _*$                                                                    | dual norm, $\max_{\ \mathbf{x}\  \leq 1} \langle \mathbf{x}, \mathbf{y} \rangle$                                    | Definition 17.4.4                        |
| $\ \mathbf{A}\ $ , $\ \mathbf{A}\ _{(k)}$                                             | unitarily invariant norm; the Ky Fan $k$ -norm<br>$\sigma_1 + \cdots + \sigma_k$                                    | Definition 20.4.1,<br>Definition 20.4.2  |
| $\Phi$ , $\Phi^*$                                                                     | symmetric gauge function and its dual, with<br>$\ \mathbf{A}\  = \Phi(\sigma(\mathbf{A}))$                          | Definition 20.3.1,<br>Definition 20.3.3  |
| $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}}$ , $\ \mathbf{x}\ _{\mathbf{A}}$ | the inner product $\mathbf{y}^* \mathbf{A} \mathbf{x}$ of a positive definite $\mathbf{A}$ ,<br>and its energy norm | Theorem 12.4.2,<br>Definition 23.7.1     |
| $\Theta(U, W)$ ,<br>$\theta_1 \leq \cdots \leq \theta_k$                              | principal angles between two subspaces, in<br>$[0, \pi/2]$                                                          | Definition 19.8.1                        |

## 23.-1.7 Bilinear forms, tensors and exterior algebra

| Symbol                                                                                 | Meaning                                                                                                                                                             | Defined at                              |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| $\beta: V \times V \rightarrow F$ ,<br>$q(\mathbf{v}) = \beta(\mathbf{v}, \mathbf{v})$ | bilinear form; the quadratic form it carries                                                                                                                        | Definition 13.1.1,<br>Definition 13.3.2 |
| $[\beta]_{\mathcal{B}}$                                                                | Gram matrix of a form, $\beta(\mathbf{v}_i, \mathbf{v}_j)$ , indices<br>uncrossed; $\mathcal{B}$ is written twice because each slot<br>of the form is given a basis | Definition 13.1.2                       |
| $\text{rad}(\beta)$                                                                    | radical of a form; $\beta$ is non-degenerate exactly<br>when it is $\{\mathbf{0}\}$                                                                                 | Definition 13.1.3,<br>Definition 13.1.4 |
| discriminant                                                                           | the square class of $\det[\beta]_{\mathcal{B}}$ in $F^\times$ , the same for<br>every basis                                                                         | Definition 13.2.2                       |
| $(n_+, n_-, n_0)$                                                                      | inertia of a real symmetric form; the signature is<br>the pair $(n_+, n_-)$                                                                                         | Definition 13.5.1                       |
| $\text{In}(\mathbf{A})$                                                                | inertia triple of a Hermitian matrix, by sign of<br>eigenvalue                                                                                                      | Definition 16.10.1                      |

| Symbol                                                                                  | Meaning                                                                                         | Defined at                                |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------|
| $\mathbf{H}_f(\mathbf{a})$                                                              | Hessian matrix of a twice-differentiable function at a point                                    | Definition 13.6.2                         |
| isotropic, totally isotropic                                                            | $q(\mathbf{v}) = 0$ with $\mathbf{v} \neq \mathbf{0}$ ; a subspace on which $\beta$ vanishes    | Definition 13.7.1,<br>Definition 13.7.2   |
| hyperbolic plane, $\text{ind}(\beta)$                                                   | the plane of a hyperbolic pair; the Witt index                                                  | Definition 13.7.3,<br>Definition 13.7.4   |
| $U \perp W$                                                                             | orthogonal direct sum for a form: the summands pair to zero                                     | Definition 13.7.3                         |
| $(V, \beta) \cong (V', \beta')$                                                         | <b>isometric</b> bilinear spaces (this use of $\cong$ is local to forms)                        | Definition 13.8.1                         |
| $\text{Isom}(\beta)$                                                                    | isometry group of a form, $\{\mathbf{P} : \mathbf{P}^\top \mathbf{A} \mathbf{P} = \mathbf{A}\}$ | Definition 13.10.1                        |
| $\text{O}(p, q)$ , $\mathbf{I}_{p, q}$ ,<br>$\text{Sp}(2m, F)$ , $\mathbf{\Omega}_{2m}$ | the indefinite orthogonal and symplectic groups and their standard matrices                     | Definition 13.10.2                        |
| $V \otimes W$ , $\mathbf{v} \otimes \mathbf{w}$                                         | tensor product and a simple tensor                                                              | Definition 14.2.1,<br>Definition 14.2.2   |
| $S \otimes T$ , $\mathcal{B} \otimes \mathcal{C}$                                       | tensor product of maps; the product basis, first factor slow                                    | Definition 14.3.1,<br>Definition 14.3.2   |
| $\Theta$                                                                                | the isomorphism $V^* \otimes W \rightarrow \mathcal{L}(V, W)$                                   | Theorem 14.3.9                            |
| $C$ , $C_b^a$                                                                           | contraction $V^* \otimes V \rightarrow F$ ; contraction of one upper against one lower slot     | Definition 14.4.1,<br>Definition 14.4.3   |
| $T_q^p(V)$ , $T_{j_1 \dots j_q}^{i_1 \dots i_p}$                                        | mixed tensors of type $(p, q)$ and their components                                             | Definition 14.4.2,<br>Definition 14.5.1   |
| $V^{\otimes k}$ , $\text{T}(V)$                                                         | $k$ -th tensor power, with $V^{\otimes 0} = F$ ; the tensor algebra                             | Definition 14.6.1,<br>Definition 14.6.2   |
| $\text{Sym}^k V$ , $\text{Sym}(V)$                                                      | $k$ -th symmetric power; the symmetric algebra, written by juxtaposition                        | Definition 14.7.2,<br>Definition 14.7.3   |
| $(V^{\otimes k})^{S_k}$                                                                 | the symmetric tensors, fixed by every permutation of the slots                                  | Definition 14.7.5                         |
| $F[x_1, \dots, x_n]$                                                                    | the polynomial algebra in $n$ variables                                                         | Definition 14.7.4                         |
| $\Lambda^k V$ , $\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_k$                         | $k$ -th exterior power and a wedge                                                              | Definition 14.8.2                         |
| $\Lambda V$ , $\mathbf{e}_S$                                                            | exterior algebra; its basis element for an increasing index set                                 | Definition 14.10.1                        |
| $\Lambda^k T$                                                                           | exterior power of a linear map; on the top power it is $\det T$                                 | Definition 14.9.1                         |
| $\varepsilon_{\mathbf{v}}$                                                              | wedging by a fixed vector, $\omega \mapsto \mathbf{v} \wedge \omega$                            | Lemma 14.9.1                              |
| $\mathcal{I}$ , $\mathcal{I}_q$                                                         | a two-sided ideal of an algebra; the one whose quotient is the Clifford algebra                 | Definition 14.10.3,<br>Definition 13.12.2 |
| $\mathbb{H}$                                                                            | the quaternions                                                                                 | Definition 13.12.3                        |

## 23.-1.8 Spectra and canonical forms

| Symbol                           | Meaning                          | Defined at                              |
|----------------------------------|----------------------------------|-----------------------------------------|
| $p_T(x) = \det(x\mathbf{I} - T)$ | characteristic polynomial, monic | Definition 6.10.1,<br>Definition 6.10.2 |

| Symbol                                                                                           | Meaning                                                                                                                                                                             | Defined at                           |
|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| $m_T(x)$                                                                                         | minimal polynomial: the monic generator of the annihilating ideal                                                                                                                   | Definition 8.5.1, Definition 5.7.1   |
| $\text{spec}(T)$                                                                                 | spectrum, the set of eigenvalues                                                                                                                                                    | Definition 8.2.3                     |
| $E_\lambda(T) = \ker(T - \lambda I)$                                                             | eigenspace                                                                                                                                                                          | Definition 8.2.1, Definition 8.2.2   |
| $a_T(\lambda), g_T(\lambda)$                                                                     | algebraic and geometric multiplicity of $\lambda$                                                                                                                                   | Definition 8.3.1, Definition 8.3.2   |
| $G_\lambda(T)$                                                                                   | generalized eigenspace, $\ker(T - \lambda I)^n$                                                                                                                                     | Definition 9.1.1, Definition 9.1.2   |
| $\mathbf{J}_k(\lambda)$                                                                          | $k \times k$ Jordan block: $\lambda$ on the diagonal, 1 on the <b>super</b> diagonal                                                                                                | Definition 9.3.1, Definition 9.3.2   |
| $\mathbf{\Lambda}(\lambda), \mathbf{C}_k(\lambda)$                                               | rotation-scaling block of a conjugate pair; the real Jordan block built from it                                                                                                     | Definition 9.4.1                     |
| $\mathbf{C}(p)$                                                                                  | companion matrix of a monic $p$                                                                                                                                                     | Definition 9.5.3                     |
| $Z(\mathbf{v}; T), m_{T, \mathbf{v}}$                                                            | cyclic subspace generated by $\mathbf{v}$ ; the $T$ -annihilator of $\mathbf{v}$                                                                                                    | Definition 9.5.1, Definition 9.5.2   |
| $d_1 \mid d_2 \mid \cdots \mid d_r$                                                              | invariant factors, with $d_r = m_T$ ; their prime powers are the elementary divisors                                                                                                | Definition 9.6.1                     |
| $\rho(\mathbf{A})$                                                                               | spectral radius, the largest modulus in the spectrum                                                                                                                                | Definition 15.4.1                    |
| $\Pi$                                                                                            | the limit $\lim_m \mathbf{A}^m$ of a convergent matrix                                                                                                                              | Theorem 9.10.4                       |
| $\mathbf{A} \succeq 0, \mathbf{A} \succ 0$                                                       | positive semidefinite, positive definite; Hermitian is implied                                                                                                                      | Definition 12.1.1                    |
| $\mathbf{A} \succeq \mathbf{B}$                                                                  | the Loewner order, $\mathbf{A} - \mathbf{B} \succeq 0$                                                                                                                              | Definition 12.5.1                    |
| $\mathbf{A}^{1/2}$                                                                               | the unique positive semidefinite square root                                                                                                                                        | Theorem 12.2.1                       |
| $ \mathbf{A}  = (\mathbf{A}^* \mathbf{A})^{1/2}$                                                 | absolute value of a matrix                                                                                                                                                          | Definition 12.2.1                    |
| $\sigma_1(\mathbf{A}) \geq \cdots \geq \sigma_{\min(m,n)}(\mathbf{A})$                           | singular values, decreasing                                                                                                                                                         | Definition 12.8.1                    |
| $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^*$                                           | a singular value decomposition: $\mathbf{U}$ and $\mathbf{V}$ unitary, $\mathbf{\Sigma}$ the same size as $\mathbf{A}$ with the singular values on its diagonal and zeros elsewhere | Theorem 12.8.1                       |
| $\mathbf{A}^+$                                                                                   | Moore–Penrose pseudoinverse                                                                                                                                                         | Definition 12.11.1                   |
| $\lambda_1(\mathbf{A}) \geq \cdots \geq \lambda_n(\mathbf{A}), \boldsymbol{\lambda}(\mathbf{A})$ | eigenvalues of a Hermitian matrix, decreasing; the vector of them                                                                                                                   | Definition 16.1.1, Definition 16.8.1 |
| $R_{\mathbf{A}}(\mathbf{x}), R_T(\mathbf{v})$                                                    | Rayleigh quotient $\langle \mathbf{A} \mathbf{x}, \mathbf{x} \rangle / \langle \mathbf{x}, \mathbf{x} \rangle$                                                                      | Definition 16.1.1                    |
| $s_k(\mathbf{A})$                                                                                | Ky Fan partial sum $\lambda_1(\mathbf{A}) + \cdots + \lambda_k(\mathbf{A})$                                                                                                         | Corollary 16.6.5                     |
| $\mathbf{d}(\mathbf{A})$                                                                         | the diagonal vector $(a_{11}, \dots, a_{nn})$ of a Hermitian matrix                                                                                                                 | Definition 16.8.1                    |
| $\mathcal{W} = (W_1, \dots, W_k)$                                                                | a flag of subspaces, and a list adapted to it                                                                                                                                       | Definition 16.7.1                    |
| $\mathcal{H}(\mathbf{A})$                                                                        | Hermitian dilation, whose eigenvalues are $\pm \sigma_i(\mathbf{A})$ and zeros                                                                                                      | Proposition 16.9.3                   |
| $r_i(\mathbf{A}), D_i(\mathbf{A})$                                                               | Gershgorin radius and disc                                                                                                                                                          | Definition 19.1.1                    |
| $O_{ij}(\mathbf{A})$                                                                             | Brauer oval of Cassini                                                                                                                                                              | Theorem 19.2.1                       |
| $W(\mathbf{A})$                                                                                  | numerical range, $\{\mathbf{x}^* \mathbf{A} \mathbf{x} : \ \mathbf{x}\ _2 = 1\}$                                                                                                    | Definition 19.2.1                    |
| $\mathbf{E}, \tilde{\mathbf{A}}$                                                                 | a perturbation, in $\mathbf{A} + \mathbf{E}$ ; a perturbed matrix when it is not of that shape                                                                                      | Definition 15.8.1                    |
| $\text{sep}_F(\mathbf{A}, \mathbf{B})$                                                           | separation of two matrices in the Frobenius norm                                                                                                                                    | Definition 19.7.1                    |

| Symbol                                | Meaning                                                                                                 | Defined at         |
|---------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------|
| $\kappa_{\mathbf{A}}(\lambda)$        | condition number of a simple eigenvalue                                                                 | Definition 19.11.1 |
| $\mathcal{S}_{\mathbf{A},\mathbf{B}}$ | Sylvester operator $\mathbf{X} \mapsto \mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B}$                     | Definition 11.10.1 |
| $\mathbf{B} - \lambda\mathbf{A}$      | a matrix pencil; its generalized eigenvalues solve $\mathbf{B}\mathbf{x} = \lambda\mathbf{A}\mathbf{x}$ | Definition 12.4.1  |

## 23.-1.9 Orders, convexity and non-negative matrices

| Symbol                                                                              | Meaning                                                                           | Defined at                                              |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------|
| $\mathbf{x} \geq \mathbf{0}, \mathbf{A} \geq 0, \mathbf{A} > 0$                     | the <b>entrywise</b> order; a positive matrix has every entry positive            | Definition 18.1.1                                       |
| $\mathbb{R}_{\geq 0}^m$                                                             | the non-negative orthant                                                          | Definition 17.5.1                                       |
| $\mathbf{x} \prec \mathbf{y}, \mathbf{x} \prec_w \mathbf{y}, \mathbf{x}^\downarrow$ | majorization, weak majorization, decreasing rearrangement                         | Definition 16.8.1                                       |
| $\text{conv } S, \text{aff } S$                                                     | convex hull; affine hull, a coset of a subspace                                   | Definition 17.1.4, Definition 17.1.5                    |
| $[\mathbf{x}, \mathbf{y}]$                                                          | the segment from $\mathbf{x}$ to $\mathbf{y}$                                     | Definition 17.1.1                                       |
| $\Delta_n$                                                                          | standard simplex: $n$ non-negative weights adding to 1                            | Definition 17.1.2                                       |
| $\text{int}(S), \overline{S}$                                                       | interior and closure in a finite-dimensional real space                           | Definition 17.3.2, Definition 17.1.6                    |
| $P_C(\mathbf{x}), d(\mathbf{x}, C)$                                                 | nearest point of a closed convex set, and the distance to it                      | Theorem 17.3.1                                          |
| $p_K, h_K, K^\circ$                                                                 | gauge, support function and polar set of a convex set                             | Definition 17.4.1, Definition 17.4.2, Definition 17.4.3 |
| $K^*$                                                                               | dual cone, inside $V$ itself                                                      | Definition 17.5.3                                       |
| $\text{cone}(\mathbf{a}_1, \dots, \mathbf{a}_m)$                                    | finitely generated cone                                                           | Definition 17.5.2                                       |
| $\text{epi } f$                                                                     | epigraph, $\{(\mathbf{x}, s) : s \geq f(\mathbf{x})\}$                            | Definition 17.10.1                                      |
| $\text{ext } C$                                                                     | extreme points of a convex set                                                    | Definition 17.8.1, Definition 17.8.2                    |
| $\mathcal{D}_n, \Omega_n$                                                           | real density matrices; doubly stochastic matrices                                 | Definition 17.8.3, Definition 17.8.4                    |
| $v(\mathbf{A})$                                                                     | value of the matrix game with payoff matrix $\mathbf{A}$                          | Definition 17.7.2, Definition 17.7.1                    |
| $\mathbf{A} \# \mathbf{B}$                                                          | geometric mean of two positive definite matrices                                  | Definition 20.11.1                                      |
| $L_f(t_1, \dots, t_n)$                                                              | Loewner matrix of divided differences                                             | Definition 20.9.2, Definition 20.9.1                    |
| $G(\mathbf{A})$                                                                     | directed graph of a square matrix: an edge $j \rightarrow i$ when $a_{ij} \neq 0$ | Definition 18.2.1                                       |
| irreducible, primitive                                                              | no proper invariant coordinate subspace; some power is positive                   | Definition 18.2.3, Definition 18.5.1                    |
| $h, \omega = e^{2\pi i/h}$                                                          | period of an irreducible non-negative matrix, and the root of unity it names      | Definition 18.5.2                                       |

| Symbol                                                         | Meaning                                                                                                                                                                    | Defined at                           |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| $r_{\mathbf{A}}(\mathbf{x}), \bar{r}_{\mathbf{A}}(\mathbf{x})$ | lower and upper Collatz–Wielandt functions; the bars are not conjugation                                                                                                   | Definition 18.4.1                    |
| Perron root and vectors                                        | $\rho(\mathbf{A})$ , the right vector $\mathbf{v} > \mathbf{0}$ with $\mathbf{1}^\top \mathbf{v} = 1$ , the left vector $\mathbf{w}$ with $\mathbf{w}^\top \mathbf{v} = 1$ | Definition 18.1.2, Definition 18.3.1 |
| $\mathbf{Q}, \mathbf{R}, \mathbf{N}$                           | transient block, absorbing block and fundamental matrix of an absorbing chain                                                                                              | Definition 18.6.1                    |
| $\mathbf{G}, \mathbf{H}, \alpha$                               | Google matrix, raw link matrix, damping factor                                                                                                                             | Definition 18.8.2, Definition 18.8.1 |
| $\mathbf{K}_n$                                                 | the second-difference matrix, a non-singular M-matrix                                                                                                                      | Definition 18.9.2, Definition 18.9.1 |
| stochastic matrix, steady state                                | column sums 1 and non-negative entries; a fixed probability vector                                                                                                         | Definition 8.11.1, Definition 8.11.2 |

## 23.-1.10 Algebras, groups and representations

| Symbol                                   | Meaning                                                                                 | Defined at                            |
|------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------|
| $A, 1_A$                                 | an associative algebra over $F$ and its unit; plain italic, never bold                  | Definition 13.12.1                    |
| $\mathcal{T}_n, \mathcal{N}$             | the upper triangular matrices in $M_n(F)$ ; its ideal of strictly upper triangular ones | Example (Six subalgebras)             |
| $F\langle S \rangle$                     | the subalgebra generated by a set: the span of the words in $S$                         | Definition 22.1.4                     |
| $F[T], F[\mathbf{A}]$                    | the algebra of polynomials in one operator or matrix, of dimension $\deg m_T$           | Proposition 22.1.2                    |
| $F[G], \delta_g$                         | group algebra of a finite group and its basis of point masses                           | Definition 22.1.5                     |
| $L_a$                                    | left multiplication by $a$ ; the regular representation is $a \mapsto L_a$              | Theorem 22.1.3                        |
| $S', S'', Z(A)$                          | commutant, double commutant, center                                                     | Definition 22.1.6, Proposition 22.1.5 |
| $C_j, L_W, R_U$                          | the column ideals of $M_n(F)$ , and the left and right ideals of a subspace             | Definition 22.2.1, Proposition 22.2.4 |
| $A/\mathcal{I}, \pi$                     | quotient algebra by a two-sided ideal, and the quotient homomorphism                    | Definition 22.2.2                     |
| $\text{rad } A$                          | radical of an algebra: killed by every irreducible representation                       | Definition 22.3.5                     |
| $a\mathbf{v}$ , an $A$ -space            | an action of $A$ on $V$ , that is a homomorphism $A \rightarrow \text{End}(V)$          | Definition 22.3.2                     |
| simple, semisimple                       | an $A$ -space with no proper non-zero invariant subspace; a direct sum of such          | Definition 22.3.3, Definition 22.3.4  |
| $\text{End}_A(V)$ , $\text{Hom}_A(V, W)$ | equivariant maps, $f(a\mathbf{v}) = af(\mathbf{v})$ ; the subscript is never dropped    | Definition 22.4.1                     |
| $A(\mathcal{F})$                         | the subalgebra generated by a family of operators                                       | Lemma 22.4.2                          |
| $\mathbf{v}\varphi$                      | the rank-one operator $\mathbf{x} \mapsto \varphi(\mathbf{x})\mathbf{v}$                | Theorem 22.5.2                        |

| Symbol                                    | Meaning                                                                          | Defined at                              |
|-------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------|
| $e_i$                                     | orthogonal idempotents of a product of algebras:<br>italic, never bold           | Definition 22.6.1                       |
| $\rho, \chi_\rho(g) = \text{tr } \rho(g)$ | a representation of a group and its character                                    | Definition 22.7.1,<br>Definition 22.8.2 |
| irreducible,<br>equivalent                | no proper non-zero subrepresentation; conjugate<br>by one isomorphism            | Definition 22.7.4,<br>Definition 22.7.3 |
| $\text{Cl}(G), \langle f, f' \rangle$     | class functions on $G$ , and their averaged inner<br>product                     | Definition 22.8.3,<br>Definition 22.8.4 |
| $V^G$                                     | the fixed space of a representation                                              | Lemma 22.8.4                            |
| $\langle g \rangle, G \times H$           | the cyclic subgroup generated by $g$ ; a direct<br>product of groups             | Definition 22.9.1,<br>Definition 22.9.2 |
| $\widehat{G}, \widehat{f}, \text{ev}_g$   | dual group of a finite abelian group, Fourier<br>transform, evaluation at $g$    | Definition 22.9.3,<br>Definition 22.9.5 |
| $\mathbf{U}$                              | character matrix, $(\mathbf{U})_{mj} = \chi_j(g_m)/\sqrt{n}$ , indices<br>from 0 | Definition 22.9.4                       |
| $\mathbf{C}, \mathbf{F}$                  | circulant matrix; the Fourier matrix                                             | Definition 11.9.1,<br>Theorem 11.9.3    |
| $\mathbf{c} * \mathbf{d}$                 | cyclic convolution, subscripts read modulo $n$                                   | Definition 11.9.1                       |

## 23.-1.11 Geometry

| Symbol                                              | Meaning                                                                                                                                     | Defined at                                                    |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| $\mathcal{A}, \mathcal{B}, \mathcal{C}$             | affine spaces, plain calligraphic                                                                                                           | Definition 21.1.1                                             |
| $P, Q, R, \overrightarrow{PQ},$<br>$P + \mathbf{v}$ | points of an affine space, <b>not bold</b> ; the vector<br>between two points; translation of a point                                       | Definition 21.1.1                                             |
| $\mathbb{A}^n(F)$                                   | affine $n$ -space: $F^n$ with its origin forgotten                                                                                          | Definition 21.1.1                                             |
| $\theta_O$                                          | the choice-of-origin bijection $\mathcal{A} \rightarrow V, P \mapsto \overrightarrow{OP}$                                                   | Proposition 21.1.3                                            |
| $\vec{f}, \widehat{f}$                              | linear part of an affine map; its block matrix,<br>homogenizing coordinate <b>last</b>                                                      | Theorem 21.1.5,<br>Proposition 21.1.7                         |
| $\text{Aff}(\mathcal{A}), \tau_{\mathbf{v}}$        | the affine group, and translation by $\mathbf{v}$                                                                                           | Theorem 21.1.6                                                |
| $\vec{A}, A \vee B, A \parallel B$                  | direction space of a flat; the join; parallelism                                                                                            | Definition 21.2.1,<br>Definition 21.2.2,<br>Definition 21.2.3 |
| $P_J$                                               | the face of a polyhedron cut out by making the<br>rows $i \in J$ tight                                                                      | Proposition 21.3.1,<br>Definition 21.3.1                      |
| $\text{E}(n), t_{\mathbf{b}}, \text{Fix}(f)$        | the Euclidean group, translation by $\mathbf{b}$ , fixed set<br>of a motion                                                                 | Definition 21.4.1                                             |
| $\mathbf{R}_\theta, \mathbf{M}_\theta$              | plane rotation and plane reflection matrices                                                                                                | Theorem 10.7.6                                                |
| $\widetilde{\mathbf{A}}, \widetilde{\mathbf{x}}$    | big matrix and big vector of a quadric, with<br>$Q(\mathbf{x}) = \widetilde{\mathbf{x}}^\top \widetilde{\mathbf{A}} \widetilde{\mathbf{x}}$ | Definition 21.5.1                                             |
| $\mathbb{P}(V), \mathbb{P}^n(F)$                    | projective space of $V$ , of dimension $\dim V - 1$ ;<br>$\mathbb{P}(F^{n+1})$                                                              | Definition 21.6.1                                             |
| $[\mathbf{x}] = [x_0 : \cdots : x_n]$               | homogeneous coordinates; coordinates run from<br>0                                                                                          | Definition 21.6.2                                             |

| Symbol                                                     | Meaning                                                                                      | Defined at                             |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------|
| $U_0, U_j, H_\infty, \iota$                                | the standard affine chart, the other charts, the hyperplane at infinity, the chart bijection | Theorem 21.6.2                         |
| $S \vee T, S \cap T$                                       | join and meet of projective subspaces                                                        | Proposition 21.6.5, Definition 21.6.3  |
| $[T], [\mathbf{A}]$                                        | the projectivity induced by an isomorphism or an invertible matrix                           | Definition 21.7.1                      |
| $\mathrm{PGL}(V), F^\times$                                | projective linear group; the multiplicative group of the field                               | Theorem 21.7.3                         |
| $(A, B; C, D)$                                             | cross-ratio of four points: itself a point of $\mathbb{P}^1(F)$ , not a scalar               | Definition 21.7.3                      |
| $S^0$                                                      | projective annihilator, Chapter 4's $U^0$ read in $\mathbb{P}(V^*)$                          | Theorem 21.8.3, Definition 21.8.1      |
| $\mathbf{o}$                                               | a chosen representative vector of a projective point                                         | Lemma 21.9.1                           |
| $Z(q), P^\beta$                                            | zero set of a quadratic form; the polar of a point with respect to a conic                   | Definition 21.10.1, Definition 21.10.2 |
| $p_{ij}, [L], \mathcal{K}$                                 | Plücker coordinates of a line, the point they form, and the Klein quadric                    | Definition 21.10.3                     |
| vol                                                        | volume: of a parallelepiped it is $ \det $ ; extended to $k$ -dimensional volume             | Definition 6.9.1, Definition 12.3.1    |
| $(\mathrm{In} \mathbf{A}, \mathrm{In} \tilde{\mathbf{A}})$ | the pair of inertia triples of a real quadric, an invariant of its affine equivalence class  | Definition 21.5.2, Definition 16.10.1  |

## 23.-1.12 Computation

| Symbol                                                             | Meaning                                                                                                                     | Defined at                             |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| $u, \mathrm{fl}, \mathcal{R}, \delta$                              | unit roundoff, the rounding map, the representable numbers, one rounding                                                    | Definition 23.1.1                      |
| $\gamma_k = ku/(1 - ku)$                                           | the accumulated-rounding constant                                                                                           | Lemma 23.1.1                           |
| $\hat{\mathbf{x}}, \Delta \mathbf{A}$                              | a computed object; a data perturbation produced by an algorithm                                                             | Definition 23.1.2, Definition 23.1.3   |
| flop, $g_n$                                                        | one arithmetic operation; the growth factor of elimination                                                                  | Definition 23.2.1, Definition 23.13.1  |
| $\mathcal{K}_k(\mathbf{A}, \mathbf{b}), \theta_1, \dots, \theta_k$ | Krylov space $\mathrm{span}(\mathbf{b}, \mathbf{A}\mathbf{b}, \dots, \mathbf{A}^{k-1}\mathbf{b})$ ; the Ritz values from it | Definition 23.6.1, Definition 23.6.2   |
| $\mathbb{L}, \mathbb{L}_0, \mu_2(\mathbb{L})$                      | graph Laplacian $\mathbf{D} - \mathbf{A}$ , the reduced Laplacian, the Fiedler value                                        | Definition 23.11.1, Definition 23.11.2 |
| $\mathbf{X}, \mathbf{X}_c, \mathbf{S}$                             | data matrix, its centered form, the sample covariance                                                                       | Definition 23.10.1, Definition 23.10.2 |
| $\mathbf{M}, \mathbf{K}, \omega_j$                                 | mass matrix, stiffness matrix, natural frequencies                                                                          | Definition 23.9.2                      |
| $\mathbf{W}_n, \mathbf{P}_n, \mathbf{D}_m, \mathbf{B}_n$           | unnormalized transform matrix, even-odd permutation, twiddle matrix, butterfly                                              | Definition 23.12.1, Definition 23.12.2 |
| $\omega$                                                           | the exponent of matrix multiplication                                                                                       | Definition 23.13.2                     |

## 23.-1.13 Where the book overrides itself

Twenty-four chapters do not have twenty-four alphabets. Several letters carry more than one meaning, and a few English words carry more than one definition. In every case the local meaning is declared where it is used, and in every case there is a tell — an argument, a subscript, a typeface — that decides which one is meant. The rows below list the clashes worth knowing before you meet them.

| Symbol or word                       | The meanings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | How to tell                                                                                                                                                                        |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>U</b>                             | a generic upper triangular matrix (Definition 0.9.5, Definition 2.6.1); a unitary matrix, in the Schur and singular value factorizations (Corollary 11.1.3, Definition 12.8.1); the character matrix of a finite abelian group (Definition 22.9.4)                                                                                                                                                                                                                                                                                                   | the chapter: triangular in Chapters 2 and 22, unitary from Chapter 11 on, the character matrix only in Chapter 22 §09                                                              |
| <b>V</b>                             | the right factor of a singular value decomposition, $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$ (Theorem 12.8.1); the Vandermonde matrix $\mathbf{V}(x_1, \dots, x_n)$ (Theorem 6.8.1); one of the two matrices invertible over $F[x]$ in $\mathbf{B} = \mathbf{UAV}$ (Definition 9.7.2). Plain $V$ is a vector space                                                                                                                                                                                                                       | the Vandermonde matrix always carries its nodes; the typeface for the space, and Chapter 9 §07 is the one place <b>V</b> is invertible only over $F[x]$                            |
| <b>irreducible</b>                   | a polynomial with no non-trivial factorization (Definition 5.3.1); a non-negative matrix with no invariant coordinate subspace (Definition 18.2.3); an $A$ -space or a representation with no proper non-zero invariant subspace (Definition 22.3.3, Definition 22.7.4)                                                                                                                                                                                                                                                                              | what the word is applied to. The three notions are unrelated                                                                                                                       |
| $e_i$ versus <b><math>e_i</math></b> | italic $e_i$ is an idempotent of a product of algebras (Definition 22.6.1); bold <b><math>e_i</math></b> is the $i$ -th standard basis vector (Example (Standard Bases))                                                                                                                                                                                                                                                                                                                                                                             | the typeface, and that is the whole reason Chapter 22 chose italic                                                                                                                 |
| <b>G</b>                             | Gram matrix (Definition 10.2.3); Givens rotation $\mathbf{G}(i, j; c, s)$ (Definition 10.8.2); Google matrix (Definition 18.8.2); the alternating Gram matrix of the Plücker form (Definition 21.10.3). Plain $G(\mathbf{A})$ is the directed graph of a matrix (Definition 18.2.1)                                                                                                                                                                                                                                                                  | the Givens rotation always carries its four arguments; the graph is plain italic with its matrix in brackets                                                                       |
| <b>H, K</b>                          | Hessian $\mathbf{H}_f(\mathbf{a})$ (Definition 13.6.2); Householder reflection $\mathbf{H}_w$ (Definition 10.8.1); the Hermitian and skew parts of a matrix (Lemma 19.2.3); the raw link matrix (Definition 18.8.1); the second-difference matrix $\mathbf{K}_n$ (Definition 18.9.2); the stiffness matrix (Definition 23.9.2). Calligraphic $\mathcal{K}_k(\mathbf{A}, \mathbf{b})$ is a Krylov space (Definition 23.6.1) and $\mathcal{K}$ the Klein quadric (Definition 21.10.3); blackboard $\mathbb{H}$ is the quaternions (Definition 13.12.3) | the Hessian and the Householder reflection always carry their argument, the second-difference matrix always its size $\mathbf{K}_n$ , the Krylov space its subscript and arguments |

| Symbol or word            | The meanings                                                                                                                                                                                                                                                                                                                                             | How to tell                                                                                                                                       |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathbb{L}$              | the lower triangular factor of an elimination (Definition 2.6.1); the graph Laplacian (Definition 23.11.1); the Lax matrix (Definition 23.8.1). Plain $L_a$ is left multiplication in an algebra (Theorem 22.1.3) and $L_f$ a Loewner matrix (Definition 20.9.2); calligraphic $\mathcal{L}(V, W)$ is a space of maps (Definition 3.4.1)                 | Chapter 23 §§08 and 11 each declare their use at the head of the section; the plain letters carry a subscript                                     |
| $\mathbf{R}, r$           | the $\mathbf{R}$ of a $QR$ factorization; the plane rotation $\mathbf{R}_\theta$ (Theorem 10.7.6); the rotation through $2\pi/n$ in Chapter 22; the absorbing block of a Markov chain (Definition 18.6.1). Plain $R_{\mathbf{A}}(\mathbf{x})$ is the Rayleigh quotient (Definition 16.1.1) and $r_i(\mathbf{A})$ a Gershgorin radius (Definition 19.1.1) | the Rayleigh quotient is plain italic and takes a vector; the Gershgorin radius is plain and takes an index                                       |
| $\mathbf{Q}$              | the orthogonal factor of a $QR$ factorization; the orthogonal linear part of a Euclidean motion (Definition 21.4.1); the quarter turn in Chapter 22; the transient block of an absorbing chain (Definition 18.6.1). Blackboard $\mathbb{Q}$ is the rationals (Definition 0.3.1)                                                                          | the chapter, and the blackboard face for the field                                                                                                |
| $\mathbf{S}, \mathcal{S}$ | a congruence or a similarity matrix; a Schur complement in Chapter 23 §02; the sample covariance (Definition 23.10.2); the cyclic shift in Chapter 11 §09. Calligraphic $\mathcal{S}_{\mathbf{A}, \mathbf{B}}$ is the Sylvester operator (Definition 11.10.1)                                                                                            | each section of Chapter 23 says which it means; the Sylvester operator carries both its matrices                                                  |
| $\mathbf{N}, \mathcal{N}$ | the fundamental matrix $(\mathbf{I} - \mathbf{Q})^{-1}$ of an absorbing chain (Definition 18.6.1); an incidence matrix, with $\mathbb{L} = \mathbf{N}\mathbf{N}^\top$ (Definition 2.7.1); the strictly upper triangular matrices $\mathcal{N}$ (Example (Six subalgebras)). Blackboard $\mathbb{N}$ is the natural numbers                               | the calligraphic face for the algebra, the blackboard face for the number set                                                                     |
| $\mathbf{J}$              | the all-ones matrix (Example (Orthogonally diagonalizing a 3 by 3 matrix)); the Jordan block $\mathbf{J}_k(\lambda)$ (Definition 9.3.1)                                                                                                                                                                                                                  | the Jordan block always carries its size and its eigenvalue                                                                                       |
| $\mathbf{C}$              | the companion matrix $\mathbf{C}(p)$ (Definition 9.5.3); the real Jordan block $\mathbf{C}_k(\lambda)$ (Definition 9.4.1); a circulant (Definition 11.9.1). Plain $C$ and $C_b^a$ are contractions (Definition 14.4.1) and $C_{ij}$ a cofactor (Definition 6.6.1); blackboard $\mathbb{C}$ is the complex numbers                                        | the companion matrix always carries its polynomial, the Jordan blocks their size and eigenvalue; a bare $\mathbf{C}$ in Chapter 22 is a circulant |
| $\mathbf{W}, W$           | the Wielandt matrix $\mathbf{W}_n$ (Theorem 18.5.6); the unnormalized transform matrix $\mathbf{W}_n$ of Chapter 23 §12; the accumulated orthogonal factor $\mathbf{W}_k$ of Chapter 23 §08. Plain $W$ is a subspace, $W(\mathbf{A})$ the numerical range (Definition 19.2.1), and $W_n$ the zero-sum matrices of Chapter 18 §07 (Lemma 18.7.1)          | the numerical range always carries its matrix as an argument; the subspace never carries one                                                      |

| Symbol or word   | The meanings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | How to tell                                                                                                                                                                                             |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\Theta, \theta$ | the isomorphism $\mathbb{C} \rightarrow \mathbb{R}^2$ (Example (The Standard Isomorphisms)); the basis-dependent $V \rightarrow V^*$ (Theorem 4.3.2); the natural isomorphism $V^* \otimes W \rightarrow \mathcal{L}(V, W)$ (Theorem 14.3.9); the diagonal matrix of principal angles (Definition 19.8.1). Lowercase: the angle between vectors (Definition 10.1.5), the choice-of-origin map $\theta_O$ (Proposition 21.1.3), the Ritz values $\theta_1, \dots, \theta_k$ (Definition 23.6.2) | the principal-angle $\Theta$ appears only inside $\sin \Theta$ and $\cos \Theta$ ; lowercase $\theta$ is an angle unless the section says otherwise, and the choice-of-origin map carries the point $O$ |
| $\Pi, \pi$       | bold $\Pi$ is the limit of a convergent stochastic matrix (Theorem 9.10.4); plain $\Pi$ is a plane in Chapter 21 §§04, 08, 09 and the quotient map $\mathrm{GL}(V) \rightarrow \mathrm{PGL}(V)$ in §07; $\pi$ is the quotient map $X \rightarrow X/\sim$ of a set (Definition 0.5.5), a quotient homomorphism of algebras (Definition 22.2.2) and, locally, a slot projection                                                                                                                  | the typeface, and that no section of Chapter 21 uses both meanings                                                                                                                                      |
| $\Phi, \Psi$     | a symmetric gauge function and its dual (Definition 20.3.1, Definition 20.3.3); the two isomorphisms of Chapter 22 §09 (Definition 22.9.5)                                                                                                                                                                                                                                                                                                                                                     | the argument: a gauge eats a real vector, Chapter 22's maps eat an element of a group algebra                                                                                                           |
| $\Omega, \omega$ | plain $\Omega_n$ is the doubly stochastic matrices (Definition 17.8.4), bold $\Omega_{2m}$ the standard alternating matrix (Definition 13.10.2). Lowercase $\omega$ is a root of unity (Definition 18.5.2, Definition 23.12.1), an angular frequency (Definition 23.9.2), the exponent of matrix multiplication (Definition 23.13.2), and a differential form being wedged (Definition 14.10.1)                                                                                                | the typeface for $\Omega$ ; for $\omega$ , the section, each of which declares its use                                                                                                                  |
| $\rho$           | spectral radius (Definition 15.4.1); a representation of a group or an algebra (Definition 22.7.1)                                                                                                                                                                                                                                                                                                                                                                                             | the argument: a matrix for the radius, a group element for the representation                                                                                                                           |
| $\sigma$         | a permutation (Definition 6.3.2); a singular value $\sigma_i(\mathbf{A})$ (Definition 12.8.1)                                                                                                                                                                                                                                                                                                                                                                                                  | the singular value always carries an index and a matrix                                                                                                                                                 |
| $\lambda, \mu$   | a scalar or an eigenvalue throughout; also the linear-part homomorphism $E(n) \rightarrow O(n)$ (Definition 21.4.1) and the Fiedler value $\mu_2$ (Definition 23.11.2)                                                                                                                                                                                                                                                                                                                         | Chapter 21 §04 is the one place $\lambda$ is not a scalar, and it says so                                                                                                                               |
| $ \cdot $        | modulus of a complex number (Definition 0.6.2); cardinality of a set; the matrix absolute value $(\mathbf{A}^* \mathbf{A})^{1/2}$ (Definition 12.2.1); the <b>entrywise</b> absolute value in Chapter 18 (Definition 18.1.1)                                                                                                                                                                                                                                                                   | what sits inside. Chapter 18 states its override, and the Chapter 12 meaning does not appear there                                                                                                      |
| $\Lambda$        | plain $\Lambda^k V$ is an exterior power (Definition 14.8.2); bold $\Lambda(\lambda)$ is a rotation-scaling block (Definition 9.4.1)                                                                                                                                                                                                                                                                                                                                                           | the typeface, and the block always carries its eigenvalue                                                                                                                                               |

| Symbol or word                                       | The meanings                                                                                                                                                                                                                                                                                                                                                                                                                                | How to tell                                                                                                                                                                                                                                                      |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\delta$                                             | Kronecker delta $\delta_{ij}$ (Definition 0.9.3); a point mass $\delta_g$ in a group algebra (Definition 22.1.5); a basis element $\delta_s$ of a free vector space (Definition 14.2.3); one rounding error (Definition 23.1.1)                                                                                                                                                                                                             | the Kronecker delta always carries two <b>numerical</b> subscripts                                                                                                                                                                                               |
| $\perp$                                              | orthogonal complement $U^\perp$ (Definition 10.3.1); the complement $U^{\perp_\beta}$ for a bilinear form (Definition 13.1.3); orthogonal direct sum $U \perp W$ of a form (Definition 13.7.3); a polarity $S^\perp$ in projective space (Theorem 21.8.3)                                                                                                                                                                                   | the unadorned $U^\perp$ is always the inner-product one                                                                                                                                                                                                          |
| $\cong, \sim, \simeq, \approx$                       | isomorphic spaces (Definition 3.5.3) or, between spaces carrying forms, isometric (Definition 13.8.1); similar matrices (Definition 3.7.2); congruent matrices (Definition 13.2.1); equivalent over $F[x]$ (Definition 9.7.2), or “approximately equal” in prose                                                                                                                                                                            | what stands on either side; the four relation symbols are never interchanged                                                                                                                                                                                     |
| brackets                                             | $[\mathbf{v}]_\mathcal{B}$ a coordinate vector (Definition 1.5.2); $[T]_\mathcal{B}^\mathcal{C}$ a matrix of a map (Definition 3.6.1); $[\mathbf{A}, \mathbf{B}]$ a commutator (Definition 7.5.1); $[\mathbf{x}, \mathbf{y}]$ a segment (Definition 17.1.1); $[\mathbf{x}]$ a projective point (Definition 21.6.2) and $[T]$ a projectivity (Definition 21.7.1); $F[x]$ and $F[G]$ rings and algebras (Definition 0.8.2, Definition 22.1.5) | the number of arguments and whether they are bold: a segment takes two bold vectors, a commutator two bold matrices                                                                                                                                              |
| the superscript of $[\cdot]_\mathcal{B}^\mathcal{C}$ | for a linear map it is the <b>output</b> basis, $[T]_\mathcal{B}^\mathcal{C}$ (Definition 3.6.1); for a bilinear form it is the basis read into the <b>second slot</b> , $[\beta]_\mathcal{B}^\mathcal{C}$ (Definition 13.1.2)                                                                                                                                                                                                              | what sits inside the brackets. The two readings are the same rule — the basis on the second slot — since a map’s second slot is its output; a form on $V \times V$ always gives both slots the same basis, so its two labels agree, and neither is ever left off |
| $T$                                                  | a linear map throughout; the transpose $\mathbf{A}^\top$ ; Chebyshev polynomials $T_n$ (Definition 23.7.4); the tensor algebra $T(V)$ (Definition 14.6.2) and mixed tensors $T_q^p(V)$ (Definition 14.4.2); a $T$ -transform (Definition 20.1.1)                                                                                                                                                                                            | the tensor algebra is upright and takes a space; the Chebyshev polynomial and the $T$ -transform carry subscripts                                                                                                                                                |
| $P, p$                                               | orthogonal projection $P_U$ (Definition 10.3.2); Legendre polynomials $P_n$ (Definition 10.10.1); a point of an affine space (Definition 21.1.1); the nearest-point map $P_C$ (Theorem 17.3.1); a face $P_J$ (Proposition 21.3.1); the even-odd permutation $\mathbf{P}_n$ (Definition 23.12.2); characteristic polynomial $p_T$ (Definition 6.10.1); a gauge $p_K$ (Definition 17.4.1)                                                     | a subspace subscript means a projection, an integer subscript a polynomial; bold means a matrix                                                                                                                                                                  |

| Symbol or word              | The meanings                                                                                                                                                                                                                                                                                                                                               | How to tell                                                                                                                  |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| $Z$                         | the cyclic subspace $Z(\mathbf{v}; T)$ (Definition 9.5.1); the center $Z(A)$ of an algebra (Definition 22.1.6); the zero set $Z(q)$ of a quadratic form (Definition 21.10.1); a Z-matrix (Definition 18.9.1)                                                                                                                                               | the argument decides: a vector and an operator, an algebra, or a form                                                        |
| $\mathbf{E}$ , $E$          | the perturbation in $\mathbf{A} + \mathbf{E}$ ; a matrix unit $\mathbf{E}_{ij}$ (Example (Standard Bases)); an elementary matrix (Definition 2.4.1); an eigenspace $E_\lambda(T)$ (Definition 8.2.2); the Euclidean group $E(n)$ (Definition 21.4.1)                                                                                                       | subscripts again: two indices give a matrix unit, an eigenvalue gives an eigenspace, a bare $\mathbf{E}$ is the perturbation |
| $\mathbf{D}$ , $D$          | a diagonal matrix (Definition 0.9.4); the elementary scaling $\mathbf{D}_i(c)$ (Definition 2.4.1); the twiddle matrix $\mathbf{D}_m$ (Definition 23.12.2); the degree matrix in $\mathbb{L} = \mathbf{D} - \mathbf{A}$ (Definition 23.11.1); a Gershgorin disc $D_i(\mathbf{A})$ (Definition 19.1.1); density matrices $\mathcal{D}_n$ (Definition 17.8.3) | the Gershgorin disc is plain italic, since the bold letter is a diagonal matrix                                              |
| $\mathbf{M}$ , $M$          | a block matrix, as in the Schur complement $\mathbf{M}/\mathbf{A}$ (Definition 7.3.1); the mass matrix (Definition 23.9.2); the plane reflection $\mathbf{M}_\theta$ (Theorem 10.7.6); a minor $M_{ij}$ (Definition 6.6.1); the space $M_n(F)$ (Definition 0.9.1); an M-matrix (Definition 18.9.2)                                                         | plain italic with two indices is a minor; $M_n(F)$ always carries its size and field                                         |
| $u$                         | the unit roundoff (Definition 23.1.1); bold $\mathbf{u}$ is a vector                                                                                                                                                                                                                                                                                       | the typeface; Chapter 23 §01 warns about the pair                                                                            |
| $h$                         | the period of an irreducible non-negative matrix (Definition 18.5.2); a support function $h_K$ (Definition 17.4.2); a mesh width in Chapter 18 §09                                                                                                                                                                                                         | the support function always carries its convex set                                                                           |
| <b>simple</b>               | a simple algebra (Definition 22.3.1); a simple $A$ -space (Definition 22.3.3); a simple eigenvalue, of multiplicity one (Definition 19.11.1); a simple tensor (Definition 14.2.2)                                                                                                                                                                          | what the word is applied to                                                                                                  |
| <b>equivalent</b>           | equivalent matrices $\mathbf{B} = \mathbf{QAP}$ (Definition 3.8.1); equivalence over $F[x]$ (Definition 9.7.2); equivalent norms (Definition 15.2.1); equivalent representations (Definition 22.7.3); affine equivalence of quadrics (Definition 21.5.2)                                                                                                   | the objects being compared                                                                                                   |
| <b>positive</b>             | $\mathbf{A} > 0$ entrywise, every entry positive (Definition 18.1.1); $\mathbf{A} \succ 0$ positive definite (Definition 12.1.1)                                                                                                                                                                                                                           | the symbol: $>$ is entrywise, $\succ$ is the Loewner order. A positive matrix need not be positive definite, nor conversely  |
| <b>radical, contraction</b> | radical of a bilinear form (Definition 13.1.3) or of an algebra (Definition 22.3.5); contraction of tensor slots (Definition 14.4.1) or a matrix of norm at most one (Definition 20.8.1)                                                                                                                                                                   | the argument, in both cases                                                                                                  |

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| Symbol or<br>word | The meanings                                                       | How to tell                                                        |
|-------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>signature</b>  | the pair $(n_+, n_-)$ , and also the single integer<br>$n_+ - n_-$ | Definition 13.5.1 says<br>which is meant where<br>the word appears |

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